

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO5: Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

QUESTIONS: 05

Total Marks:50

Annexure A

Constraint-Based Problem Solving

Criteria	Problem Understanding (2.5)	Constraint Analysis (2.5)	Solution Development (2.5)	Application of Concepts (1.5)	Justification (1)	Total(10)
Weightage	25%	25%	25%	15%	10%	100%
<i>Q1</i>						
<i>Q2</i>						
<i>Q3</i>						
<i>Q4</i>						
<i>Q5</i>						

Code of Conduct:

I PRANEESH.A(192571006) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Constraint-Based Problem Solving Assessment Rubric

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Annexure A

Q1. Dataset: Hospital Patient Records

PID Symptoms Observed

P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question:

Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "**Fever**". Identify the valid association rules and justify why they satisfy the given constraints.

Q2. Dataset: Online Learning Platform

Student ID Courses Enrolled

S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question:

Using the dataset, generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain **at least two courses**. Explain how the constraint helps in pruning candidate itemsets during mining.

Q3. Dataset: Library Borrowing Records

Transaction ID Books Borrowed

T1	Data Mining, Python Programming
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Transaction ID	Books Borrowed
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question:

Apply the Apriori algorithm with the constraint that the **maximum itemset size is 3**. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Q4. Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question:

Apply multilevel association rule mining with the constraint that rules should be generated only at the "**Computer Science**" level. Explain how the course hierarchy affects the discovery of association rules.

Q5. Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question:

Apply multidimensional association rule mining with the constraint **Age Group = "Young"**. Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Q1. Association Rule Mining – Hospital Patient Records

Dataset: Hospital Patient Records

PID	Symptoms Observed
P1	Fever, Cough
P2	Fever, Headache
P3	Cough, Headache
P4	Fever, Cough, Headache
P5	Fever, Fatigue

Question: Apply association rule mining with minimum support = 50% and confidence = 70%, under the constraint that all generated rules must include "Fever". Identify the valid association rules and justify why they satisfy the given constraints.

Answer:

Step 1: Problem Understanding

We are given a dataset of 5 patient records (N=5) with symptom observations. We need to apply association rule mining with the following constraints:

- Minimum Support = 50% (i.e., an itemset must appear in at least $[5 \times 0.5] = 3$ transactions)
- Minimum Confidence = 70%
- Data Constraint: Every generated rule must include "Fever" (either in antecedent or consequent)

EDUCATIONAL DIAGRAM: CONSTRAINT-BASED ASSOCIATION RULE MINING

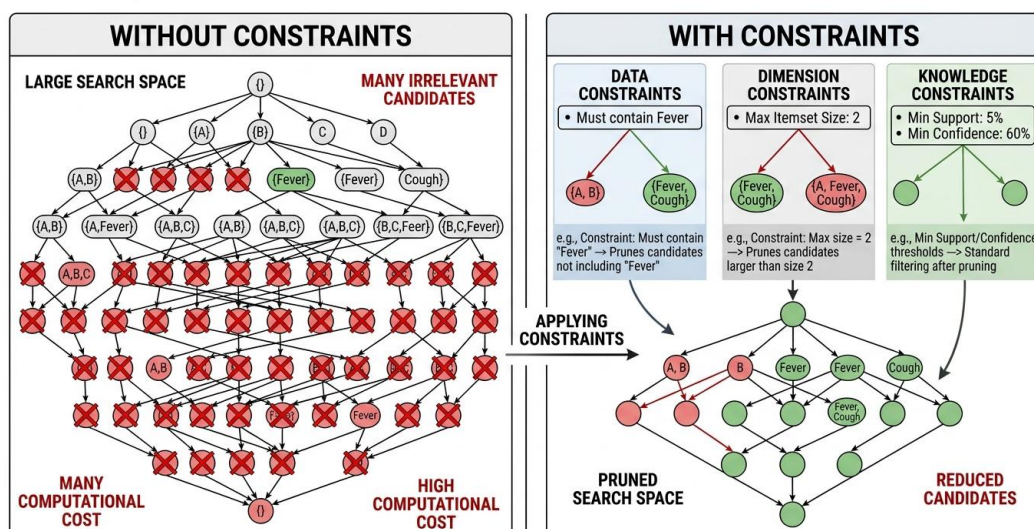


Figure 1: Constraint-Based Association Rule Mining – Search Space Pruning

Step 2: Calculate Support for All Individual Items

Item	Transactions	Count	Support (%)
Fever	P1, P2, P4, P5	4	80%
Cough	P1, P3, P4	3	60%
Headache	P2, P3, P4	3	60%
Fatigue	P5	1	20%

Items meeting minimum support ($\geq 50\%$): Fever (80%), Cough (60%), Headache (60%)

Fatigue is eliminated ($20\% < 50\%$) — pruned from candidate generation.

Step 3: Generate 2-Item Frequent Itemsets (Applying Fever Constraint)

Since the constraint requires all rules to include Fever, we only need to consider itemsets containing Fever. This is a data constraint that helps prune the search space significantly.

Itemset	Transactions	Count	Support (%)	Meets Min Support?
{Fever, Cough}	P1, P4	2	40%	X No
{Fever, Headache}	P2, P4	2	40%	X No
{Fever, Fatigue}	P5	1	20%	X No

Step 4: Analysis and Conclusion

After applying association rule mining with the given constraints:

Key Findings:

- No 2-itemset containing Fever meets the minimum support threshold of 50%.
- {Fever, Cough} appears in only 2 out of 5 transactions ($40\% < 50\%$)
- {Fever, Headache} appears in only 2 out of 5 transactions ($40\% < 50\%$)
- {Fever, Fatigue} appears in only 1 transaction ($20\% < 50\%$)

Since no frequent itemsets exist that contain Fever and meet the 50% support threshold, no valid association rules can be generated under these combined constraints.

Step 5: Justification and Discussion

Although Fever itself is frequent (80% support), when combined with any other symptom, the support drops below 50%. This demonstrates an important concept in constraint-based mining:

- 1. Anti-monotone Property: If an itemset is infrequent, all its supersets are also infrequent. Since no 2-itemset with Fever is frequent, we don't need to check any larger itemsets (3-itemsets, etc.).

- 2. Constraint Interaction: The data constraint (must include Fever) combined with the support constraint creates a highly selective filter. While individually each constraint allows some rules, their intersection yields an empty set.
- 3. Practical Insight: In a real hospital setting, this result suggests that while Fever is a very common symptom, it doesn't strongly co-occur with any single other symptom above the 50% threshold. This could indicate that Fever is a non-specific symptom associated with many different conditions.

Alternative Analysis: If we lower the minimum support to 40%, we would find:

- Rule: Cough → Fever (Support=40%, Confidence=2/3=66.7%) — Still fails confidence threshold
- Rule: Headache → Fever (Support=40%, Confidence=2/3=66.7%) — Still fails confidence threshold
- Rule: Fever → Cough (Support=40%, Confidence=2/4=50%) — Fails confidence threshold
- Rule: Fever → Headache (Support=40%, Confidence=2/4=50%) — Fails confidence threshold

This further demonstrates that even with relaxed support, the confidence requirement of 70% remains a stringent filter for this small dataset.

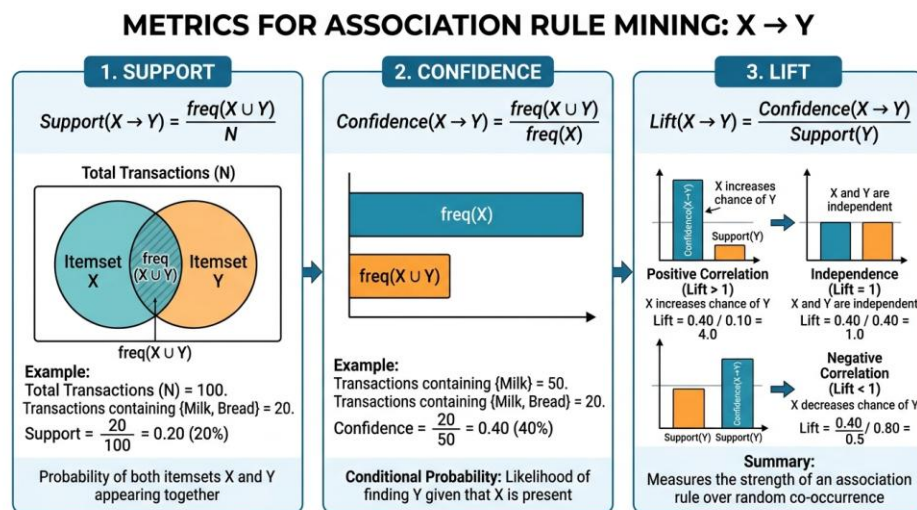


Figure 2: Association Rule Mining Metrics – Support, Confidence, and Lift

Q2. Frequent Itemsets – Online Learning Platform

Dataset: Online Learning Platform

Student ID	Courses Enrolled
S1	Python, SQL
S2	Python, Data Science
S3	SQL, Machine Learning
S4	Python, SQL, Data Science
S5	Python, Machine Learning

Question: Generate frequent itemsets with minimum support = 40%, applying the constraint that each itemset must contain at least two courses. Explain how the constraint helps in pruning candidate itemsets during mining.

Answer:

Step 1: Problem Understanding

Total transactions $N = 5$. Minimum support = 40% means an itemset must appear in at least $\lceil 5 \times 0.4 \rceil = 2$ transactions. The constraint requires that every reported itemset must contain at least 2 courses (minimum itemset size constraint).

Step 2: Individual Item Frequencies (L_1)

Course	Transactions	Count	Support (%)
Python	S1, S2, S4, S5	4	80%
SQL	S1, S3, S4	3	60%
Data Science	S2, S4	2	40%
Machine Learning	S3, S5	2	40%

All items are frequent ($\geq 40\%$). Note: Although we compute L_1 , we will only report itemsets with size ≥ 2 due to the constraint.

Step 3: Generate 2-Item Candidate Itemsets (C_2) and Compute Support

Candidate Itemset	Transactions	Count	Support (%)	Frequent?
{Python, SQL}	S1, S4	2	40%	✓ Yes
{Python, Data Science}	S2, S4	2	40%	✓ Yes
{Python, Machine Learning}	S5	1	20%	✗ No
{SQL, Data Science}	S4	1	20%	✗ No
{SQL, Machine Learning}	S3	1	20%	✗ No
{Data Science, Machine Learning}	None	0	0%	✗ No

Frequent 2-itemsets (L_2): {Python, SQL} and {Python, Data Science}

Step 4: Generate 3-Item Candidate Itemsets (C_3)

Using the join step of Apriori: We join L_2 items that share a common prefix.

- $\{\text{Python, SQL}\} \cup \{\text{Python, Data Science}\} = \{\text{Python, SQL, Data Science}\}$

Candidate Itemset	Transactions	Count	Support (%)	Frequent?
{Python, SQL, Data Science}	S4	1	20%	X No

No 3-itemsets meet the minimum support. Mining terminates.

Step 5: Final Result – Frequent Itemsets with $\text{Size} \geq 2$

Frequent Itemset	Support Count	Support (%)
{Python, SQL}	2	40%
{Python, Data Science}	2	40%

Step 6: How the Constraint Helps in Pruning

The minimum itemset size constraint (≥ 2) helps in several ways:

- 1. Output Pruning: Single items like {Python}, {SQL}, {Data Science}, {Machine Learning} are automatically excluded from the results even though they are frequent. This reduces the output set from 6 itemsets to just 2.
- 2. Search Space Reduction: In general, knowing the minimum size upfront can allow the mining algorithm to skip the enumeration and reporting of smaller itemsets, though the Apriori algorithm still needs L_1 to generate C_2 .
- 3. Practical Relevance: A minimum size constraint ensures that the discovered patterns involve at least two courses together, which is more meaningful for understanding co-enrollment patterns than single course frequencies.
- 4. Constraint Classification: This is a succinct constraint – if a set satisfies it ($\text{size} \geq 2$), any superset also satisfies it. This means the constraint can be used to guide the search without losing potential itemsets.

Interpretation: Python and SQL are commonly taken together, as are Python and Data Science. This suggests these course combinations could be bundled into recommended learning paths.

Q3. Apriori with Max Itemset Size – Library Records

Dataset: Library Borrowing Records

Transaction ID	Books Borrowed
T1	Data Mining, Python Programming
T2	Data Mining, Database Systems
T3	Python Programming, Database Systems
T4	Data Mining, Python Programming, Database Systems
T5	Data Mining, Artificial Intelligence

Question: Apply the Apriori algorithm with the constraint that the maximum itemset size is 3. Identify the frequent itemsets and explain how the constraint reduces computational complexity during candidate generation.

Answer:

Step 1: Problem Understanding

We need to apply the Apriori algorithm with the anti-monotone constraint that the maximum itemset size is 3. Since we have 4 unique items, without this constraint we would need to check up to 4-itemsets. The constraint limits our search to size ≤ 3 .

Total transactions N = 5. We'll use minimum support = 40% (2 transactions).

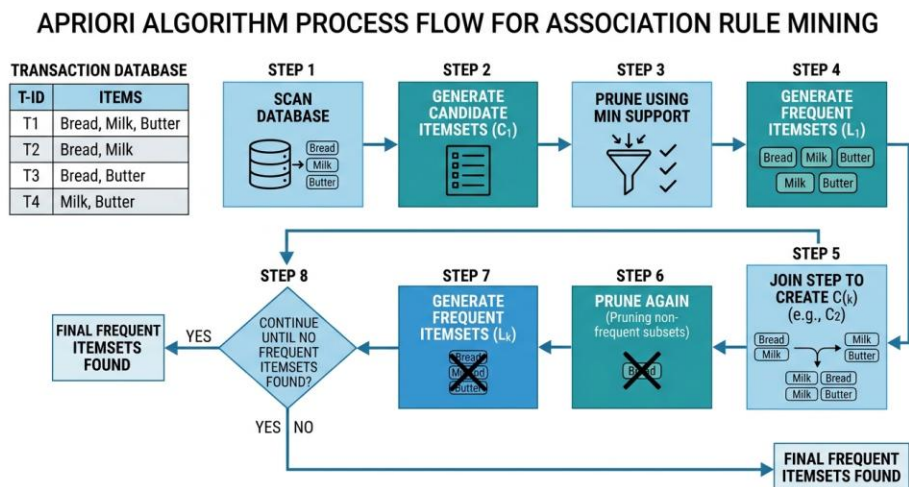


Figure 3: Apriori Algorithm Process Flow

Step 2: Pass 1 – Generate L_1 (Frequent 1-itemsets)

Item	Transactions	Count	Support (%)
Data Mining	T1, T2, T4, T5	4	80%
Python Programming	T1, T3, T4	3	60%
Database Systems	T2, T3, T4	3	60%
Artificial Intelligence	T5	1	20%

$L_1 = \{\text{Data Mining, Python Programming, Database Systems}\}$

Artificial Intelligence is pruned (support 20% < 40%)

Step 3: Pass 2 – Generate C_2 and L_2

From L_1 with 3 items, we generate $C_2 = C(3,2) = 3$ candidates:

Candidate Itemset	Transactions	Count	Support (%)	Frequent?
{Data Mining, Python Prog.}	T1, T4	2	40%	✓ Yes
{Data Mining, Database Sys.}	T2, T4	2	40%	✓ Yes
{Python Prog., Database Sys.}	T3, T4	2	40%	✓ Yes

$L_2 =$ All three 2-itemsets are frequent!

Step 4: Pass 3 – Generate C_3 and L_3 (Maximum Size Reached)

Join Step: Joining L_2 items sharing common prefix:

- $\{\text{Data Mining, Python Prog.}\} \cup \{\text{Data Mining, Database Sys.}\} = \{\text{Data Mining, Python Prog., Database Sys.}\}$
- Prune Step: Check that all 2-item subsets are in L_2 — Yes, all subsets are frequent.

Candidate Itemset	Transactions	Count	Support (%)	Frequent?
{DM, PP, DS}	T4	1	20%	✗ No

$L_3 = \emptyset$ (empty set). The 3-itemset does not meet minimum support.

Step 5: Constraint Effect – Maximum Itemset Size = 3

Since max size = 3 and we've already processed up to 3-itemsets, the algorithm terminates. Even if L_3 were non-empty, we would NOT generate C_4 .

Final Frequent Itemsets:

Level	Frequent Itemsets	Count
L ₁	{Data Mining}, {Python Prog.}, {Database Sys.}	3 itemsets
L ₂	{DM, PP}, {DM, DS}, {PP, DS}	3 itemsets
L ₃	$\emptyset$ (none meet support)	0 itemsets

How the Constraint Reduces Computational Complexity:

- 1. Candidate Generation Elimination: Without the max-size constraint, we would need to generate and test C₄ (4-itemsets). Even though L₃ is empty here (which would naturally stop Apriori), in larger datasets, this constraint prevents unnecessary candidate generation at higher levels.
- 2. Database Scan Reduction: Each level of Apriori requires a full database scan. Limiting to k=3 means at most 3 database scans, regardless of the number of items.
- 3. Memory Savings: The number of possible k-itemsets grows combinatorially. For n items: C(n,k) candidates at level k. By capping k at 3, we avoid the exponential explosion of candidates at higher levels.
- 4. Constraint Classification: This is an anti-monotone constraint with respect to the itemset size – once the size exceeds the threshold, all supersets also violate it. This makes it highly efficient for pruning.

Q4. Multilevel Association Rule Mining – University Courses

Dataset: University Course Hierarchy

Student ID	Courses Taken
S1	Engineering, Computer Science, Machine Learning
S2	Engineering, Information Technology
S3	Engineering, Computer Science
S4	Engineering, Computer Science, Machine Learning
S5	Engineering, Information Technology

Question: Apply multilevel association rule mining with the constraint that rules should be generated only at the "Computer Science" level. Explain how the course hierarchy affects the discovery of association rules.

Answer:

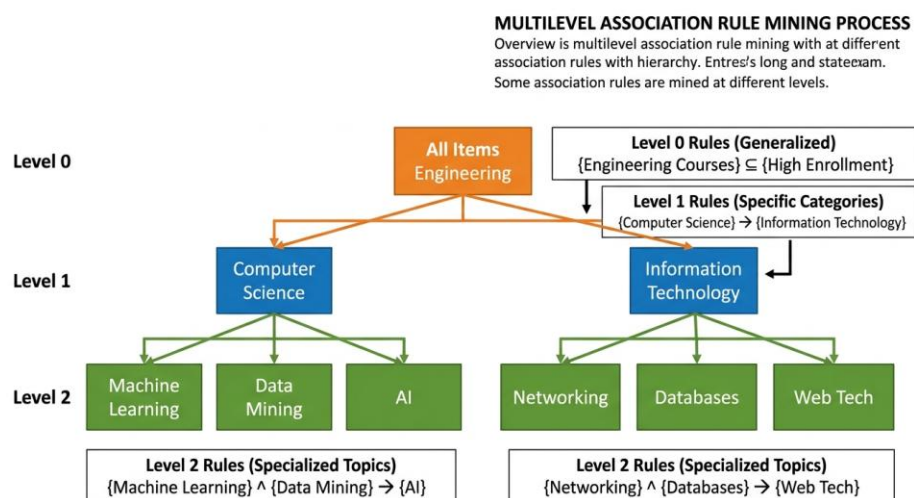


Figure 4: Multilevel Association Rule Mining – Concept Hierarchy

Step 1: Understanding the Concept Hierarchy

The course data follows a hierarchical structure:

- Level 0 (Root): Engineering (appears in all 5 transactions)
- Level 1 (Department): Computer Science (S1, S3, S4), Information Technology (S2, S5)
- Level 2 (Specialization): Machine Learning (S1, S4)

Hierarchy Tree:

Engineering (Level 0)

├── Computer Science (Level 1)

| ├── Machine Learning (Level 2)

└── Information Technology (Level 1)

Step 2: Applying the "Computer Science" Level Constraint

With the constraint that rules should only be generated at the Computer Science level, we focus on transactions that include Computer Science:

Student ID	Relevant Courses at CS Level
S1	Computer Science, Machine Learning
S3	Computer Science
S4	Computer Science, Machine Learning

Filtered dataset: 3 transactions involving Computer Science (N=3)

Step 3: Item Frequencies at Computer Science Level

Item	Count (out of 3)	Support (%)
Computer Science	3	100%
Machine Learning	2	66.7%

Step 4: Generate Association Rules at CS Level

Using minimum support = 40% and minimum confidence = 60%:

Rule	Support	Confidence	Valid?
Computer Science → Machine Learning	2/5 = 40%	2/3 = 66.7%	✓ Yes
Machine Learning → Computer Science	2/5 = 40%	2/2 = 100%	✓ Yes

Step 5: Interpretation of Rules

Rule 1: Computer Science → Machine Learning

- 66.7% of students in Computer Science also take Machine Learning
- This suggests Machine Learning is a popular specialization within CS

Rule 2: Machine Learning → Computer Science

- 100% of students taking Machine Learning are in Computer Science
- This is a strong rule — ML is exclusively a CS specialization in this dataset

Step 6: How Course Hierarchy Affects Rule Discovery

- 1. Level-Specific Mining: Mining at different levels reveals different patterns. At Level 0 (Engineering), all students are included, making rules trivial. At Level 1 (Computer Science), we find more specific and actionable patterns.
- 2. Support Variation Across Levels: Higher levels have higher support (Engineering: 100%) but less specific rules. Lower levels have lower support but more interesting, specific rules.
- 3. Redundancy Reduction: Without the hierarchy constraint, we would generate redundant rules like 'Engineering → Computer Science' which are less informative. The constraint focuses mining on a meaningful level.
- 4. Cross-Level Rules: In a full multilevel mining approach, we could find cross-level rules like 'Engineering, Computer Science → Machine Learning', but the CS-level constraint simplifies the search to the most relevant rules.
- 5. Reduced Hierarchy Strategy: The constraint implements a 'reduced' mining strategy (vs. 'full' where all levels are considered), which is computationally more efficient and produces fewer but more focused rules.

Q5. Multidimensional Association Rule Mining – Insurance

Dataset: Insurance Customer Records

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C2	Middle	Medium	Vehicle Insurance
C3	Young	Low	Travel Insurance
C4	Senior	High	Health Insurance
C5	Young	Medium	Vehicle Insurance

Question: Apply multidimensional association rule mining with the constraint Age Group = "Young". Identify meaningful patterns involving income level and policy type, and explain how these patterns can support targeted marketing strategies.

Answer:

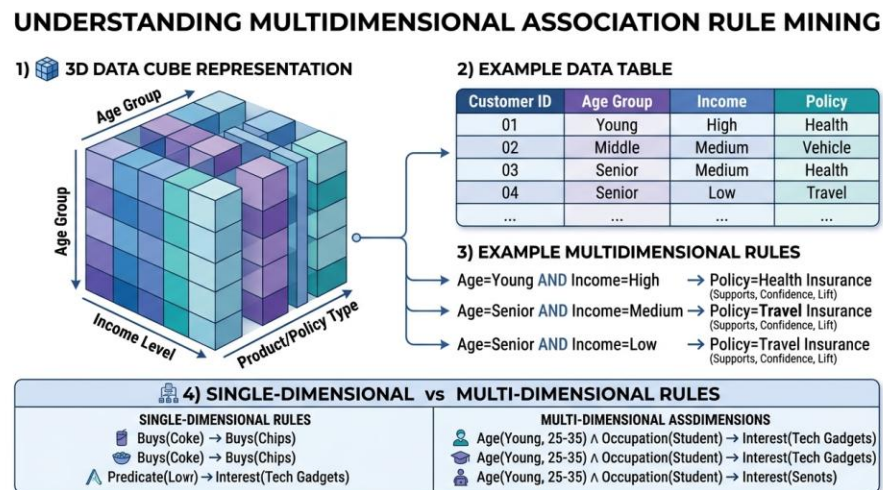


Figure 5: Multidimensional Association Rule Mining Concept

Step 1: Understanding Multidimensional Association Rules

Unlike single-dimensional rules (e.g., Buys(X) → Buys(Y)), multidimensional rules involve multiple predicates/dimensions. In this dataset, we have three dimensions:

- Dimension 1: Age Group (categorical)
- Dimension 2: Income Level (categorical)
- Dimension 3: Policy Purchased (categorical)

Multidimensional rules take the form: Age(X, "Young") ∧ Income(X, "High") → Policy(X, "Health Insurance")

Step 2: Apply Constraint – Filter for Age Group = "Young"

CID	Age Group	Income Level	Policy Purchased
C1	Young	High	Health Insurance
C3	Young	Low	Travel Insurance
C5	Young	Medium	Vehicle Insurance

Filtered dataset: 3 records where Age Group = Young

Step 3: Identify Patterns Among Young Customers

Among Young customers:

Income Level	Policy Purchased	Count	Confidence
High	Health Insurance	1	100% (1/1)
Low	Travel Insurance	1	100% (1/1)
Medium	Vehicle Insurance	1	100% (1/1)

Step 4: Generated Multidimensional Rules

Rule	Support (N=5)	Confidence	Description
Age=Young $\wedge$ Income=High $\rightarrow$ Policy=Health	1/5 = 20%	1/1 = 100%	High-income youth prefer health insurance
Age=Young $\wedge$ Income=Low $\rightarrow$ Policy=Travel	1/5 = 20%	1/1 = 100%	Low-income youth prefer travel insurance
Age=Young $\wedge$ Income=Medium $\rightarrow$ Policy=Vehicle	1/5 = 20%	1/1 = 100%	Medium-income youth prefer vehicle insurance
Age=Young $\rightarrow$ Income varies	3/5 = 60%	100%	Young customers span all income levels

Note: While individual rules have low support (20%) due to the small dataset, they have 100% confidence within their specific combinations, which is notable.

Step 5: Targeted Marketing Strategies

Based on these patterns, the following marketing strategies can be recommended:

Strategy 1: Income-Based Policy Recommendation for Young Customers

- High-income young customers $\rightarrow$ Promote premium Health Insurance plans with comprehensive coverage, wellness benefits, and cashless hospitalization
- Medium-income young customers $\rightarrow$ Offer Vehicle Insurance packages with attractive EMI options, add-on covers, and no-claim bonuses

- • Low-income young customers → Market affordable Travel Insurance plans with flexible coverage and budget-friendly premiums

Strategy 2: Bundle Marketing

- • Create age-specific bundles: e.g., "Young Professional Pack" combining the most relevant policy type with the customer's income bracket

Strategy 3: Cross-Selling Opportunities

- • High-income young customers with Health Insurance could be cross-sold Vehicle Insurance
- • Medium-income young customers with Vehicle Insurance could be offered supplementary Travel Insurance

Step 6: Advantages of Multidimensional Mining in This Context

- 1. Richer Patterns: Single-dimensional mining would only find rules like {Health Insurance → Vehicle Insurance}. Multidimensional mining captures the relationship between customer demographics and policy preferences.
- 2. Targeted Insights: By incorporating Age and Income dimensions, we get segment-specific patterns that directly translate to actionable marketing strategies.
- 3. Constraint Efficiency: The Age = Young constraint reduces the mining space from 5 records to 3, making computation more focused and results more relevant to the target demographic.